

Лабораторная работа №6

Администрирование локальных сетей

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Информация

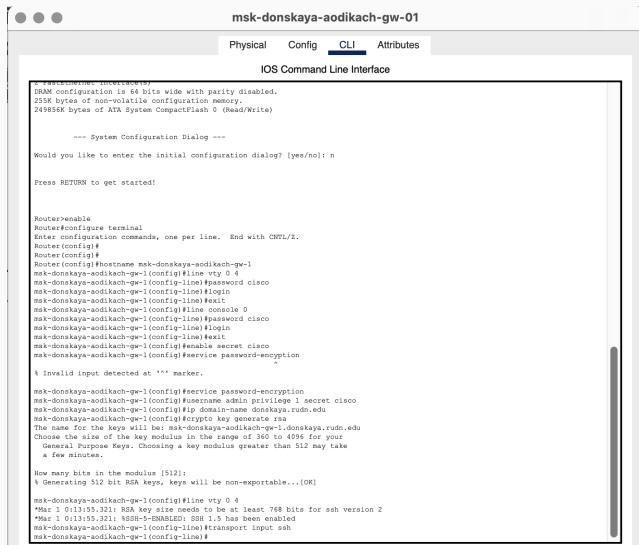
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- Российский университет дружбы народов
- <https://github.com/chelibos?tab=repositories>

Настроить статическую маршрутизацию VLAN в сети.

Выполнение лабораторной работы

Добавляю маршрутизатор Cisco 2811 в сеть и подключаю его к 24 порту коммутатора msk-donskaya-aodikch-sw-01.

Провожу первичное конфигурирование маршрутизатора. Настраиваю имя, пароль, удалённое подключение по ssh



```
msk-donskaya-aodikach-gw-01
Physical Config CLI Attributes
IOS Command Line Interface

* Cisco IOS Software (C6400) (C6400-ADVENTERK9-M)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: n

Press RETURN to get started!

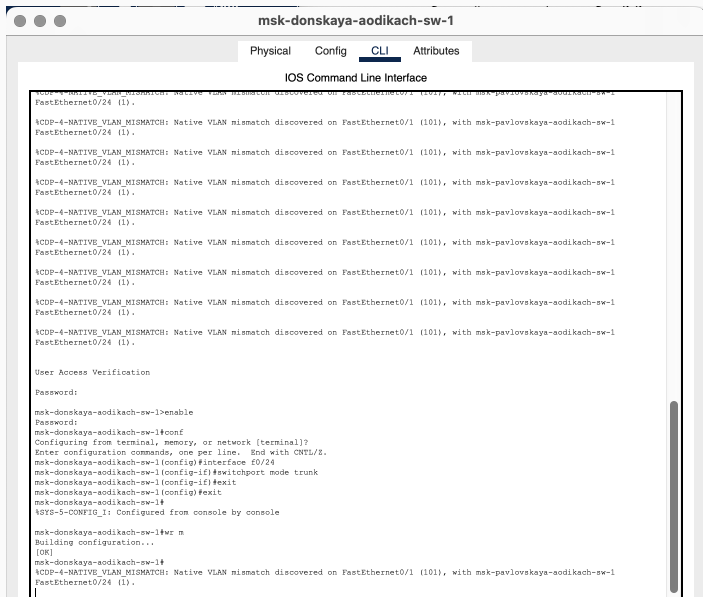
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#hostname msk-donskaya-aodikach-gw-1
msk-donskaya-aodikach-gw-1(config)#line vty 0 4
msk-donskaya-aodikach-gw-1(config-line)#password cisco
msk-donskaya-aodikach-gw-1(config-line)#login
msk-donskaya-aodikach-gw-1(config-line)#exit
msk-donskaya-aodikach-gw-1(config)#line console 0
msk-donskaya-aodikach-gw-1(config-line)#password cisco
msk-donskaya-aodikach-gw-1(config-line)#login
msk-donskaya-aodikach-gw-1(config-line)#exit
msk-donskaya-aodikach-gw-1(config)#enable secret cisco
msk-donskaya-aodikach-gw-1(config)#service password-encryption
^
% Invalid input detected at '^' marker.

msk-donskaya-aodikach-gw-1(config)#service password-encryption
msk-donskaya-aodikach-gw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-aodikach-gw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-aodikach-gw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-aodikach-gw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

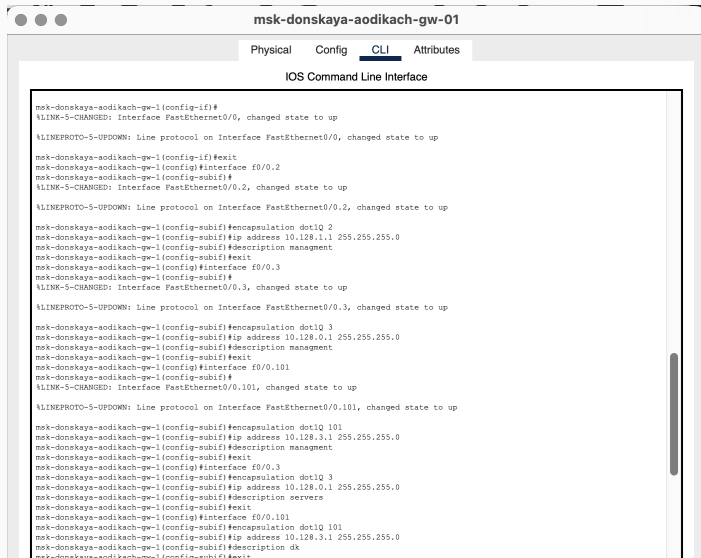
How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-aodikach-gw-1(config)#line vty 0 4
*Mar 1 0:13:55.321: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:13:55.321: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-aodikach-gw-1(config-line)#transport input ssh
msk-donskaya-aodikach-gw-1(config-line)#
```

Настраиваю порт коммутатора на msk-donskaya-aodikach-sw-01 как trunk-порт



Настраиваю виртуальные интерфейсы в соответствии с номерами VLAN опираясь на таблицу IP-адресов



```
msk-donskaya-aodikach-gw-01
Physical Config CLI Attributes
IOS Command Line Interface

msk-donskaya-aodikach-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

msk-donskaya-aodikach-gw-1(config-if)#exit
msk-donskaya-aodikach-gw-1(config)#interface f0/0.2
msk-donskaya-aodikach-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.2, changed state to up

msk-donskaya-aodikach-gw-1(config-subif)#encapsulation dot1Q 2
msk-donskaya-aodikach-gw-1(config-subif)#ip address 10.128.1.1 255.255.255.0
msk-donskaya-aodikach-gw-1(config-subif)#description managment
msk-donskaya-aodikach-gw-1(config-subif)#exit
msk-donskaya-aodikach-gw-1(config)#interface f0/0.3
msk-donskaya-aodikach-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.3, changed state to up

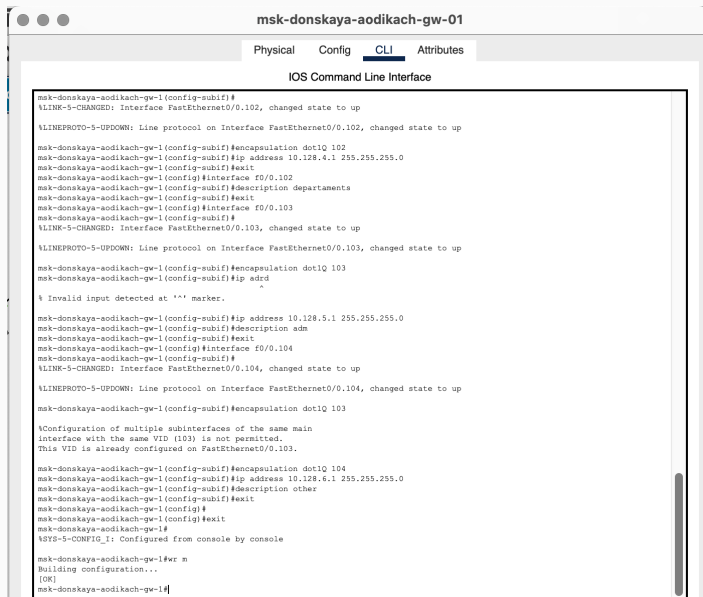
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.3, changed state to up

msk-donskaya-aodikach-gw-1(config-subif)#encapsulation dot1Q 3
msk-donskaya-aodikach-gw-1(config-subif)#ip address 10.128.0.1 255.255.255.0
msk-donskaya-aodikach-gw-1(config-subif)#description managment
msk-donskaya-aodikach-gw-1(config-subif)#exit
msk-donskaya-aodikach-gw-1(config)#interface f0/0.101
msk-donskaya-aodikach-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.101, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.101, changed state to up

msk-donskaya-aodikach-gw-1(config-subif)#encapsulation dot1Q 101
msk-donskaya-aodikach-gw-1(config-subif)#ip address 10.128.3.1 255.255.255.0
msk-donskaya-aodikach-gw-1(config-subif)#description managment
msk-donskaya-aodikach-gw-1(config-subif)#exit
msk-donskaya-aodikach-gw-1(config)#interface f0/0.3
msk-donskaya-aodikach-gw-1(config-subif)#encapsulation dot1Q 3
msk-donskaya-aodikach-gw-1(config-subif)#ip address 10.128.0.1 255.255.255.0
msk-donskaya-aodikach-gw-1(config-subif)#description servers
msk-donskaya-aodikach-gw-1(config-subif)#exit
msk-donskaya-aodikach-gw-1(config)#interface f0/0.101
msk-donskaya-aodikach-gw-1(config-subif)#encapsulation dot1Q 101
msk-donskaya-aodikach-gw-1(config-subif)#ip address 10.128.3.1 255.255.255.0
msk-donskaya-aodikach-gw-1(config-subif)#description dk
msk-donskaya-aodikach-gw-1(config-subif)#exit
```

Продолжение настройки портов



```
msk-donskaya-aodikach-gw-01
Physical Config CLI Attributes
IOS Command Line Interface

msk-donskaya-aodikach-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.102, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.102, changed state to up

msk-donskaya-aodikach-gw-1(config-subif)#encapsulation dot1Q 102
msk-donskaya-aodikach-gw-1(config-subif)#ip address 10.128.4.1 255.255.255.0
msk-donskaya-aodikach-gw-1(config-subif)#exit
msk-donskaya-aodikach-gw-1(config)#interface f0/0.102
msk-donskaya-aodikach-gw-1(config-subif)#description departaments
msk-donskaya-aodikach-gw-1(config-subif)#exit
msk-donskaya-aodikach-gw-1(config)#interface f0/0.103
msk-donskaya-aodikach-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.103, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.103, changed state to up

msk-donskaya-aodikach-gw-1(config-subif)#encapsulation dot1Q 103
msk-donskaya-aodikach-gw-1(config-subif)#ip adr
^
% Invalid input detected at '^' marker.

msk-donskaya-aodikach-gw-1(config-subif)#ip address 10.128.5.1 255.255.255.0
msk-donskaya-aodikach-gw-1(config-subif)#description adm
msk-donskaya-aodikach-gw-1(config-subif)#exit
msk-donskaya-aodikach-gw-1(config)#interface f0/0.104
msk-donskaya-aodikach-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.104, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.104, changed state to up

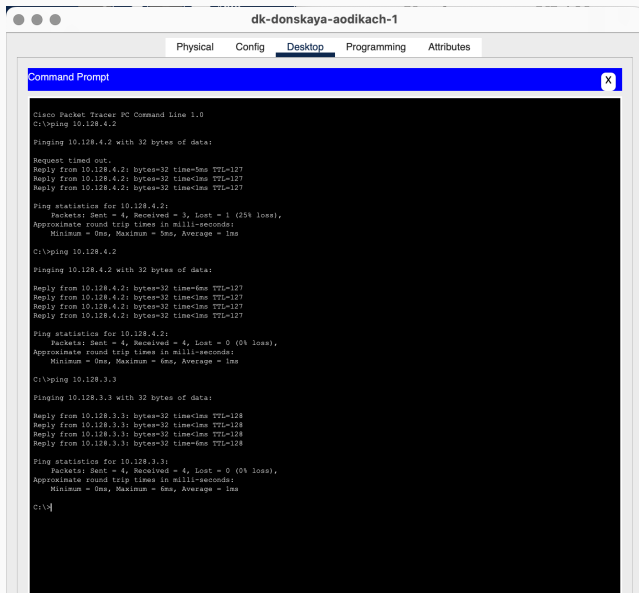
msk-donskaya-aodikach-gw-1(config-subif)#encapsulation dot1Q 103

%Configuration of multiple subinterfaces of the same main
interface with the same VID (103) is not permitted.
This VID is already configured on FastEthernet0/0.103.

msk-donskaya-aodikach-gw-1(config-subif)#encapsulation dot1Q 104
msk-donskaya-aodikach-gw-1(config-subif)#ip address 10.128.6.1 255.255.255.0
msk-donskaya-aodikach-gw-1(config-subif)#description other
msk-donskaya-aodikach-gw-1(config-subif)#exit
msk-donskaya-aodikach-gw-1(config)#
msk-donskaya-aodikach-gw-1(config)#exit
msk-donskaya-aodikach-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-aodikach-gw-1#wr m
Building configuration...
[OK]
msk-donskaya-aodikach-gw-1#
```

Проверяю доступность устройств с помощью команды ping



The screenshot shows a desktop environment titled "dk-donskaya-aodikach-1" with tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a "Command Prompt" window. The window contains the output of a Cisco Packet Tracer PC Command Line 1.0 session, showing two successful ping tests to 10.128.4.2 and 10.128.3.3.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.4.2

Pinging 10.128.4.2 with 32 bytes of data:

Request timed out.
Reply from 10.128.4.2: bytes=32 time=5ms TTL=127
Reply from 10.128.4.2: bytes=32 time<1ms TTL=127
Reply from 10.128.4.2: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.4.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 1ms

C:\>ping 10.128.4.2

Pinging 10.128.4.2 with 32 bytes of data:

Reply from 10.128.4.2: bytes=32 time=6ms TTL=127
Reply from 10.128.4.2: bytes=32 time<1ms TTL=127
Reply from 10.128.4.2: bytes=32 time<1ms TTL=127
Reply from 10.128.4.2: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.4.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 1ms

C:\>ping 10.128.3.3

Pinging 10.128.3.3 with 32 bytes of data:

Reply from 10.128.3.3: bytes=32 time<1ms TTL=128
Reply from 10.128.3.3: bytes=32 time<1ms TTL=128
Reply from 10.128.3.3: bytes=32 time<1ms TTL=128
Reply from 10.128.3.3: bytes=32 time=6ms TTL=128

Ping statistics for 10.128.3.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 1ms

C:\>
```

Запускаю симуляцию и просматриваю процесс передвижения пакета ICMP

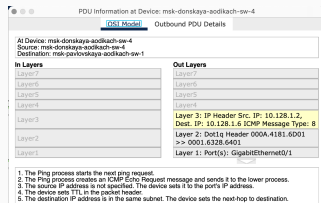


Рис. 6: Симуляция

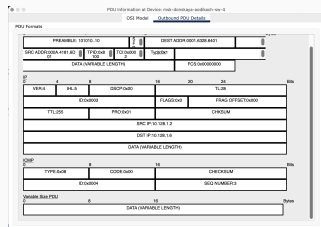


Рис. 7: Симуляция

Настроила статистическую маршрутизацию VLAN в сети.